Understanding the lung cancer classification portfolio project

A research guide to the data methods results and scientific limitations

Project study guide | 24 October 2026

Abstract

This project asks whether genes supported by several feature selection methods can distinguish lung adenocarcinoma from lung squamous cell carcinoma while retaining an interpretable biological signal. It implements an adapted reproduction of Chen and Dhahbi (2021), then adds normal tissue classification, model interpretation, survival analysis and candidate discovery. The saved analysis contains 1,029 tumour samples, 110 normal samples and 15,686 retained genes. Five selectors generate a 339 gene panel supported by at least two methods and a 140 gene panel supported by at least three.

On the saved binary test split, a random forest using the 339 gene panel correctly classifies 299 of 309 samples, with accuracy 0.9676 and AUROC 0.9887. Its advantage over several alternative panels is one correctly classified sample. Individual genes also discriminate the subtypes in an independent microarray cohort, but the complete multigene classifier has not been externally validated. Important limitations include preprocessing before splitting, feature selection outside inner cross validation, reuse of the test set across analyses, and weaknesses in the three class split. Survival findings are exploratory. The strongest conclusion is that the project demonstrates a reproducible analysis workflow and strong subtype associated expression; it does not establish a clinically validated diagnostic or prognostic assay.

How to read this guide

Read Chapters 1 to 3 for the biological question and data preparation, Chapters 4 to 8 for the models and results, and Chapters 9 to 11 for the validity audit, reproduction plan and manuscript framing. Chapter 12 provides definitions and an evidence map. Tables use saved result files; the full pipeline was not rerun to write this guide.

The original manuscript style report is useful for following the analysis. This guide adds explanations and a stricter reading of what its results can support. Historical file labels such as overlap131 and overlap17 are retained in the code, even though their actual panels contain 339 and 140 genes.

1 Research question and study design

Lung adenocarcinoma (LUAD) and lung squamous cell carcinoma (LUSC) are distinct tumour histologies. This project studies whether their differences in gene expression can be used for classification and interpreted biologically. Each sample is represented by a vector of expression measurements: one value per gene. The binary target is the known histology label. Normal tissue is introduced later as a separate extension.

The central challenge is that there are many more genes than samples. With 15,686 candidate variables and 1,029 tumours, a model can exploit chance patterns, correlated genes or technical effects. Feature selection reduces the input to a smaller panel. The project compares five ways of selecting genes and asks whether agreement between them gives a useful panel.

Chen and Dhahbi proposed selecting genes supported by at least two methods for classification and at least three for biomarker assessment [1]. Their article supplies the conceptual design. The present implementation changes several statistical procedures and inputs, so it is best described as an adapted reproduction with extensions. Matching a published panel size is not the same as reproducing its gene identities or predictive performance.

Three questions that must remain separate

Question	What provides evidence
Can expression classify histology?	Predictions on patients excluded from model development
Which genes describe subtype biology?	Expression differences, pathway enrichment and independent replication
Do genes predict survival?	Survival models with clinical covariates and external follow up

A gene can be an excellent subtype discriminator without predicting survival. A survival association can arise from disease stage or other factors without a causal role for the gene. Likewise, a model explanation identifies what a fitted model uses; it does not establish a molecular mechanism.

For the reader, the most useful mental picture is a sequence: assemble the expression matrix, normalize it, reserve test patients, select genes using training data, train a common classifier for every panel, then inspect biological and external evidence. The implementation mostly follows that structure for the binary analysis, with the exceptions detailed in Chapter 10.

2 Cohorts labels and data provenance

Data component	Saved analysis scope	Purpose
TCGA LUAD	528 tumours	Training and binary evaluation
TCGA LUSC	501 tumours	Training and binary evaluation
TCGA normal tissue	110 samples	Three class extension
Expression matrix	15,686 genes after filtering	Common input feature universe
GSE28582 GPL570 subset	50 LUAD and 28 LUSC	External single gene discrimination

The TCGA inputs are expression matrices obtained through the UCSC Xena GDC hub. The project uses STAR count files. Its source documentation contrasts these with the HTSeq files used in the anchor study. Changes in quantification, annotation, filtering and mapping can all change the retained gene universe. The difference between 15,686 genes here and 14,010 in the source study should therefore not be attributed to annotation alone without a controlled comparison.

TCGA barcodes supply patient identity and sample type. The code treats primary tumour code 01 as tumour and solid tissue normal code 11 as normal, excluding other sample types. Histology follows the cohort from which a sample was loaded. Multiple samples from one person are not independent patients. Keeping their identity through preprocessing is essential to evaluation.

GSE28582 is a GEO SuperSeries containing more than one assay type [2]. The project reads the GPL570 expression matrix, identifies histology from sample metadata and excludes other diagnoses. Its saved documentation reports 78 eligible samples. Affymetrix probe identifiers are mapped to gene symbols; where multiple probes map to a gene, the implementation takes the maximum expression per sample. This aggregation rule is a methodological choice that can influence the external AUC.

Files and sample counts

The final portfolio snapshot includes source code, reports, figures and result tables. It omits raw expression inputs and intermediate processed matrices, which are downloaded or regenerated. Consequently, the cohort sizes above are the saved analysis counts, not a new recount from the raw files performed for this guide.

A stronger provenance record would include the exact source URL, retrieval date, checksum, annotation version and sample inclusion list for every input. The train and test patient lists should be versioned alongside the run manifest. These records make it possible to distinguish a code change from a changed upstream dataset.

3 Preparing expression data

From downloaded values to a usable matrix

The loader assumes that the downloaded Xena values are log2(count + 1). It reverses that transformation, rounds to nonnegative counts and stores them using memory efficient numeric types. This assumption is appropriate only for files documented to use that encoding; it must be checked if an input file is replaced. Applying the inverse transformation to ordinary counts or another normalized matrix would corrupt the analysis.

The LUAD and LUSC matrices are merged by gene identifier and duplicate sample columns are removed. Before gene symbol mapping, the code drops genes with fewer than one count per million in more than 600 tumour samples. Across 1,029 tumours, this means a retained gene must reach the CPM threshold in at least 429 samples. It is a fixed rule tied to this cohort size, not a general cutoff for every future study.

Ensembl version suffixes are removed before matching identifiers to a gene symbol map. For example, a versioned identifier and its unversioned base can represent the same gene. Counts sharing a symbol are summed. Filtering before symbol aggregation reduces memory use, but can differ from filtering after aggregation: two individually weak identifiers could jointly form a retained gene after summing.

What normalization achieves

Trimmed Mean of M values (TMM) estimates sample scaling factors to reduce differences in library size and RNA composition. Counts are then expressed as log2 counts per million with a prior count. This reduces scale differences and compresses highly skewed expression values. It does not automatically remove batch effects, tumour purity differences, all mean variance dependence or every technical confounder.

The current pipeline calculates the low count filter and TMM transformation before reserving test samples. No outcome labels are needed for those steps, but the test distribution still contributes to the representation. A stricter deployment style evaluation would define the retained genes and normalization reference using training data, then apply a prespecified transformation to new samples.

What to inspect before modeling

Check sample identities, duplicated patients, missing values, sample type counts, library size distributions and the proportion of genes that fail identifier mapping. Plot sample structure and examine whether cohort, processing batch or tumour purity tracks the histology label. A clean software run does not substitute for these checks.

4 Understanding the five feature selectors

The five selectors are complementary algorithms fitted to the binary training samples. They are not independent experiments: all see the same patients and many respond to the same correlated expression signal. Agreement can prioritize robust candidates, but it does not provide five independent confirmations or guarantee control of false discoveries.

Selector	How it chooses genes	Saved panel
Differential expression	Welch tests on log CPM with BH FDR < 0.05 and absolute log fold change > 0.5; top 500 by p value	500
PCA	Standardize genes; rank absolute loadings on the first principal component	500
mRMR variant	Greedy scaled ANOVA relevance minus mean absolute Pearson correlation to selected genes	500
XGBoost	Rank positive gain importance from boosted trees and retain the top 148	148
L1 logistic regression	Choose the strongest penalty within one standard error of the best cross validated accuracy; retain nonzero coefficients	18

Differential expression asks which genes differ on average between histologies. The code uses Welch tests rather than the negative binomial model in edgeR; calling this exact edgeR reproduction would be incorrect. PCA asks which genes drive the largest standardized expression axis, without using labels during fitting. That axis may reflect histology, but this should be checked rather than assumed.

The mRMR implementation balances association with histology against similarity to genes already selected. It uses scaled F statistics and correlations, so it is a project specific relevance redundancy criterion rather than an exact mutual information based reproduction. XGBoost captures nonlinear patterns; correlated genes can share or exchange importance. L1 regression produces a sparse linear selection, but may choose one member of a correlated group somewhat arbitrarily.

Gene membership is then counted across the five lists. At least two votes produce 339 genes, at least three produce 140, and all five produce six: BCL2L15, DDAH1, DSC3, DSG3, KRT5 and SIAH2. The historical names overlap131 and overlap17 refer to the source study, not the panel sizes in this implementation. Use the actual sizes when presenting results. [P1, P2]

The source study reported 131 and 17 overlap genes. These counts depend on input data, implementation and selection depth. In the saved sensitivity analysis, selecting the top 50 from the ranked selectors gives 51 genes with at least two votes and 17 with at least three; at depth 500, those counts are 350 and 143 under that sensitivity configuration. Similar counts do not guarantee the same genes. XGBoost selection is explicitly capped at the source study size of 148.

5 Evaluation and the meaning of each metric

The binary split is stratified by histology and groups samples by patient within each histology. Approximately 70% of the tumours are assigned to training and 30% to testing. The saved confusion counts show 309 test samples: 159 LUAD and 150 LUSC, leaving 720 training samples. The split is saved once and reused across feature panels, allowing a comparison on the same test examples.

Every panel is evaluated using a random forest. This common classifier helps distinguish the value of the selected inputs from differences in the classifier itself. A grid search uses five fold cross validation within the training set to choose the number of trees, the candidate features at each split and the minimum leaf size. Random forests combine many decision trees to reduce instability relative to one tree.

Metric	Meaning in this project
Accuracy	Fraction of all test samples with the correct histology
Sensitivity	Fraction of LUAD test samples correctly classified
Specificity	Fraction of LUSC test samples correctly classified
Precision	Fraction of predicted LUAD samples that are actually LUAD
F1 score	Harmonic mean of precision and sensitivity for LUAD
AUROC	How well model scores rank LUAD above LUSC across thresholds

LUAD is the positive class. A sensitivity of 1.000 means all 159 LUAD test samples were correctly classified on this split; it does not mean perfect future detection. AUROC measures ranking rather than calibration. A model can rank samples well and still assign poorly calibrated probabilities or perform badly at a chosen operating threshold.

The saved accuracy confidence interval uses 10,000 bootstrap resamples of the fixed test predictions. For the 339 gene panel it spans approximately 94.8% to 98.7%. That interval reflects variation in the scored test sample, conditional on this fitted model. It does not include the full uncertainty from choosing a split, selecting genes, tuning the model or changing cohorts. Patient cluster bootstrapping would be preferable if test patients contribute multiple samples. [P1, P2]

6 Binary classification results

Feature panel	Genes	Accuracy	F1	AUROC
DGE	500	0.9579	0.9602	0.9862
PCA	500	0.9644	0.9664	0.9825
mRMR	500	0.9644	0.9666	0.9869
XGBoost	148	0.9644	0.9666	0.9892
Lasso	18	0.9579	0.9598	0.9913
Overlap at least 2	339	0.9676	0.9695	0.9887
Overlap at least 3	140	0.9579	0.9605	0.9846

The 339 gene panel has the highest saved accuracy and F1. It correctly classifies every LUAD sample and 140 of 150 LUSC samples. Its ten errors are LUSC samples predicted as LUAD. Precision is therefore 159 divided by 169, or approximately 94.1%. This makes the reported metrics concrete: the main error pattern is asymmetric.

PCA, mRMR and XGBoost panels each classify 298 of 309 samples correctly, compared with 299 for the overlap panel. That one sample difference is too small to establish superiority without a paired comparison and repeated independent evaluation. L1 selection produces the highest AUROC in the saved table, 0.9913, despite lower accuracy. Therefore, the statement that overlap is the best method needs a named metric and a qualification about this split. [P2]

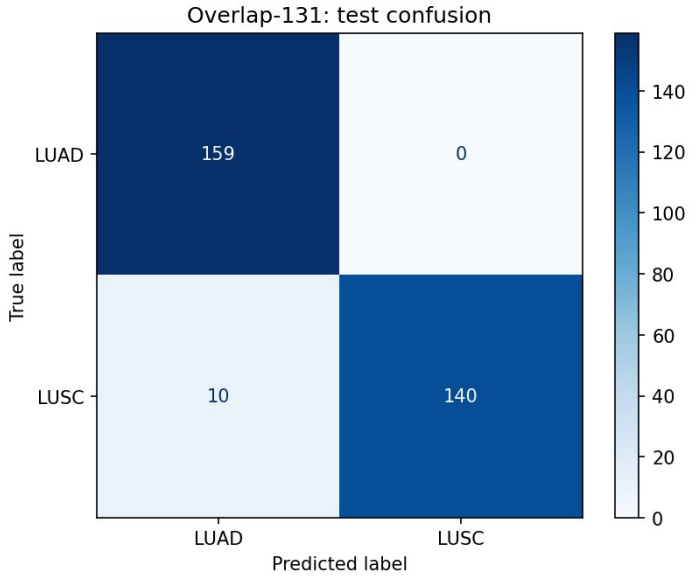


Figure 1. Saved confusion matrix for the 339 gene overlap panel. The historical figure label says overlap 131; the actual panel contains 339 genes.

7 Biomarkers and biological interpretation

The 140 genes supported by at least three selectors are evaluated one at a time. This asks whether the expression of a single gene separates the histologies. It is distinct from evaluating the 339 gene random forest. TCGA AUCs use the full tumour cohort, including samples used to select these genes, so they are descriptive discovery results rather than held out estimates. A strong single gene AUC does not show that a fixed threshold will work across laboratories or that the gene adds value beyond established markers.

In the TCGA table, DSG3 has AUC 0.9743 and KRT5 has AUC 0.9732. Both are higher in LUSC in this analysis. The emphasis on keratins and epithelial adhesion is consistent with the squamous expression program discussed in the anchor study. Such consistency helps interpret the signal, but expression differences may also reflect cellular composition and differentiation rather than tumour specific causal mechanisms.

Gene	TCGA AUC	External AUC
DSG3	0.974	0.986
KRT5	0.973	0.988
DSC3	0.969	0.993
CERS3	0.965	0.847
KRT6A	0.953	0.994

The external result table contains 134 mapped genes, with mean AUC 0.9253. This is the arithmetic mean of 134 separate gene AUCs. It is not the performance of a combined classifier, an average across 134 patients or an estimate of clinical utility. The cohort provides independent samples and a different measurement platform, which is useful evidence of biological transportability.

However, the AUC function independently uses the larger of AUC and one minus AUC in each cohort. This chooses the favorable expression direction after inspecting that cohort's labels. A stronger external test would freeze the direction from training, then retain that sign in the external cohort. Any gene whose direction reverses should be reported explicitly. No fixed multigene prediction model is transferred to GSE28582 in the current code. [P3]

Pathway analysis uses over representation tests against the analysed gene background, separately for genes up in LUSC and LUAD. Keratinization is prominent on the LUSC side. Gene set enrichment indicates that a list contains more annotated pathway members than expected; it does not directly measure pathway activation, explain every tumour or prove a treatment mechanism. [P4]

8 Extensions and what they add

Three class prediction

Adding 110 normal samples changes the target to LUAD, LUSC or normal tissue. The saved 339 gene model has accuracy 0.9591, macro F1 0.9574 and macro AUROC 0.9917. The all gene model has accuracy 0.9620 and macro F1 0.9638. Macro F1 gives equal weight to each class, which helps prevent the smaller normal class from disappearing inside a high overall accuracy.

These results are preliminary. The extension creates a new split after selecting genes on the binary split. In addition, its grouping occurs independently within each class: a patient's tumour and normal tissue can be allocated to opposite sides. This also affects the all gene baseline. One global patient assignment must govern all classes and all selection steps before the three class result can be treated as a clean estimate. [P5]

Interpretation and smaller panels

TreeSHAP explains an XGBoost model on the overlap genes. It attributes model outputs to individual features, with DSC3 and KRT5 among the reported drivers. It does not explain the random forest directly or establish causal effects. Separately, an XGBoost ranking is used to build progressively larger random forest panels. The saved ablation AUCs are 0.9546 for two genes, 0.9799 for ten, 0.9845 for seventeen and 0.9881 for twenty five. Repeatedly inspecting this test curve turns it into development evidence; a new validation set is needed after choosing a final panel size. [P6]

Candidate discovery and clustering

The exploratory stability analysis resamples training data 30 times and records genes selected by an L1 model or XGBoost in a prefiltered pool. DSG3 and CERS3 have saved selection frequencies of 1.00. This shows consistency under that resampling recipe, not a formal guarantee of biological truth or novelty. A literature review is required before describing any candidate as newly discovered.

Within histology clustering gives a silhouette of 0.303 for two LUSC clusters and 0.149 for two LUAD clusters. The LUSC structure is worth investigating, but batch, purity and other technical factors must be examined before naming new molecular subtypes. The sex audit is also limited: absence of a detected sex association does not exclude other confounders or prove equivalence. [P7]

9 Survival analysis and statistical interpretation

The survival extension tests each of 140 genes separately within each tumour histology, giving 280 gene histology comparisons. Kaplan Meier curves compare high and low expression groups split at the within histology median. Log rank tests compare the survival curves. A separate univariate Cox model uses expression as a continuous predictor and estimates a hazard ratio per expression unit. These two tests ask related but different questions.

The saved analysis reports 63 log rank p values below 0.05 without multiplicity correction. To illustrate the effect of testing many genes, this guide additionally applies the Benjamini Hochberg procedure once across all 280 saved log rank p values. Eleven comparisons remain below q = 0.05. This is a new calculation from the saved table, not a rerun of the survival models. A different prespecified testing family would yield different adjusted values. [P8]

LUAD gene	Log rank p	BH q across 280	Cox HR
PERP	0.000208	0.0109	1.356
KRT6A	0.002771	0.0560	1.135
KRT17	0.015520	0.1403	1.098

Under this adjustment, the quoted KRT6A and KRT17 log rank associations do not meet q < 0.05. Their Cox p values are separate tests and must not be substituted after seeing this result. If Cox results are used as the primary evidence, their own testing family and adjustment should be defined in advance.

The Cox models omit stage, age, sex, treatment and other clinical covariates. The code also does not report checking the proportional hazards assumption. A hazard ratio greater than one describes an association with higher event hazard under the model, not an increase in absolute risk by that proportion and not evidence that changing the gene changes survival.

The next useful analysis is a prespecified multivariable survival model with a clear endpoint, patient level records, missing data handling, event counts, proportional hazards checks and independent validation. Median splits can make plots intuitive but discard expression information; continuous models should be primary when their functional assumptions are reasonable. The present survival results generate hypotheses rather than validated prognostic claims.

10 Validity audit and stronger evaluation

The binary evaluation has a useful held out patient structure, but the workflow does not keep every data dependent decision inside the proper validation boundary. The following issues matter more than another decimal place of accuracy.

Preprocessing and cross validation

Filtering and normalization use the full expression cohort. Within the training set, feature selection is performed once before random forest cross validation. Thus each validation fold has already influenced which genes are available. The inner cross validation score can be optimistic. The outer binary test score still provides information if the test labels did not influence development, but it should not be called a completely untouched deployment style evaluation.

The L1 selector also fits its standardization once before its own cross validation. That exposes each inner validation fold's mean and variance to training. A pipeline should fit scaling within each fold. If multiple samples from a patient occur in training, ordinary sample based inner cross validation can split them across folds; group aware inner folds are needed as well. [P1]

Repeated decisions based on test performance

Seven panels, multiple thresholds and several panel sizes are scored on the same test set. Reporting the winning configuration after seeing all of them introduces selection effects. Similar accuracy over several settings is encouraging descriptive evidence, but does not demonstrate statistical equivalence or guarantee generalization. Nested patient grouped cross validation would estimate the entire selection and tuning procedure, followed by a genuinely untouched external test for the finalized model.

A defensible next study

1. Assign each patient globally to development or validation before any class specific split, and record the assignment.

2. Fit gene filtering, scaling, selection and classifier tuning inside patient grouped training folds. Separate discovery analyses from confirmatory tests.

3. Freeze the selected panel, expression direction, normalization approach and decision threshold. Test the full classifier on independent samples with compatible measurements.

4. Report class specific error rates, calibration, uncertainty across splits and the performance difference against simpler baselines. Validate survival separately with covariate adjustment.

The current project supports strong subtype discrimination in retrospective data and an interpretable computational workflow. It does not establish population screening performance, causal biomarkers, a validated clinical threshold or treatment benefit.

11 Reproducing and presenting the project

A practical reading and execution order

Begin with README.md and the executed notebooks/walkthrough.ipynb. The notebook shows the results without requiring expensive recomputation. Then inspect docs/REPORT.md for the analysis narrative, docs/MODELS.md for the implementation choices and docs/DATA.md for input provenance. Use this guide alongside those documents to distinguish the implementation from the stronger claims in parts of the original narrative.

The final portfolio copy has 130 files. The source scripts define an ordered pipeline: preprocessing; feature selection; binary classification; biomarker ROC analysis; pathway enrichment; three class classification; SHAP and ablation; survival; candidate stability and clustering; sensitivity analysis; and a summary figure. Configuration and shared helpers live in src/config.py and src/utils.py.

Stage	Entry point
Install isolated dependencies	python -m venv .venv then activate it and install requirements.txt
Check software behavior	make test
Acquire the public input files	bash download_data.sh or make data
Run the analysis	make reproduce
Review the evidence	Compare regenerated tables with results/tables and inspect plots

Download links can change. Confirm expected file types, matrix encoding, checksums and cohort counts rather than assuming the downloader's successful exit proves the correct data were obtained. The archive omits raw data, processed matrices and analysis caches. Some stages reuse cached outputs, so a reproduction record must state which stages were recomputed and which were reused. Tests on synthetic data verify software behavior; they do not independently validate the biological conclusions.

The Cowork handoff reported six passing tests and an executed notebook. Those are historical run results, not new checks of this report. Before publishing a manuscript, reproduce in a clean environment with pinned versions and preserve the run log, input manifest, split assignments and generated tables. No package installation, analysis rerun or GitHub publishing was performed to prepare this guide.

A suitable manuscript claim

An adapted overlapping feature selection workflow identified compact panels with strong retrospective discrimination between TCGA LUAD and LUSC. Cross platform single gene analyses supported several subtype associated signals. The small observed differences between panels, validation boundary limitations and exploratory survival analyses motivate stricter patient grouped and external evaluation before clinical claims.

12 Definitions references and evidence map

Term	Plain language meaning
Gene expression	Measured abundance of RNA associated with a gene
Feature	An input variable; here usually one gene
Training set	Samples used to develop the model and select genes
Test set	Reserved samples used to estimate predictive performance
Cross validation	Repeated training and evaluation within a development dataset
Overfitting	Learning patterns that do not transfer to new samples
Data leakage	Allowing evaluation data to influence development decisions
FDR and q value	Multiple testing framework; q values express adjusted significance
Calibration	Agreement between predicted probabilities and observed frequencies
Confounding	An additional factor that can explain an observed association

External references

[1] Chen JW and Dhahbi J. Lung adenocarcinoma and lung squamous cell carcinoma cancer classification, biomarker identification, and gene expression analysis using overlapping feature selection methods. Scientific Reports. 2021;11:13323. https://doi.org/10.1038/s41598-021-92725-8

[2] NCBI Gene Expression Omnibus. GSE28582 and its GPL570 expression data. https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE28582

Project evidence

Evidence identifiers refer to paths inside the recovered tcga-lung-subtype-classifier folder. [P1] src/00_preprocess.py, src/01_feature_selection.py, src/02_classification.py and src/utils.py establish preprocessing, selection and evaluation behavior. [P2] results/tables/classification_metrics.csv and feature_sets.json provide the binary numbers and panel memberships. [P3] src/03_biomarkers_roc.py and biomarker_auc_tcga.csv / biomarker_auc_external.csv establish the single gene AUC procedure and saved results.

[P4] src/04_pathways.py and results/tables/enrichment_up.csv / enrichment_down.csv support pathway interpretation. [P5] src/05a_three_class.py and threeclass_metrics.csv establish the three class procedure and results. [P6] src/05b_shap_ablation.py and ablation_auc.csv support interpretation and panel size analysis. [P7] src/05d_novel.py, novel_candidates.csv, subclusters.csv and confound_audit.csv support exploratory analyses. [P8] src/05c_survival.py and survival_biomarkers.csv support survival results; the BH adjusted values in Chapter 9 were calculated for this guide across 280 saved log rank tests.